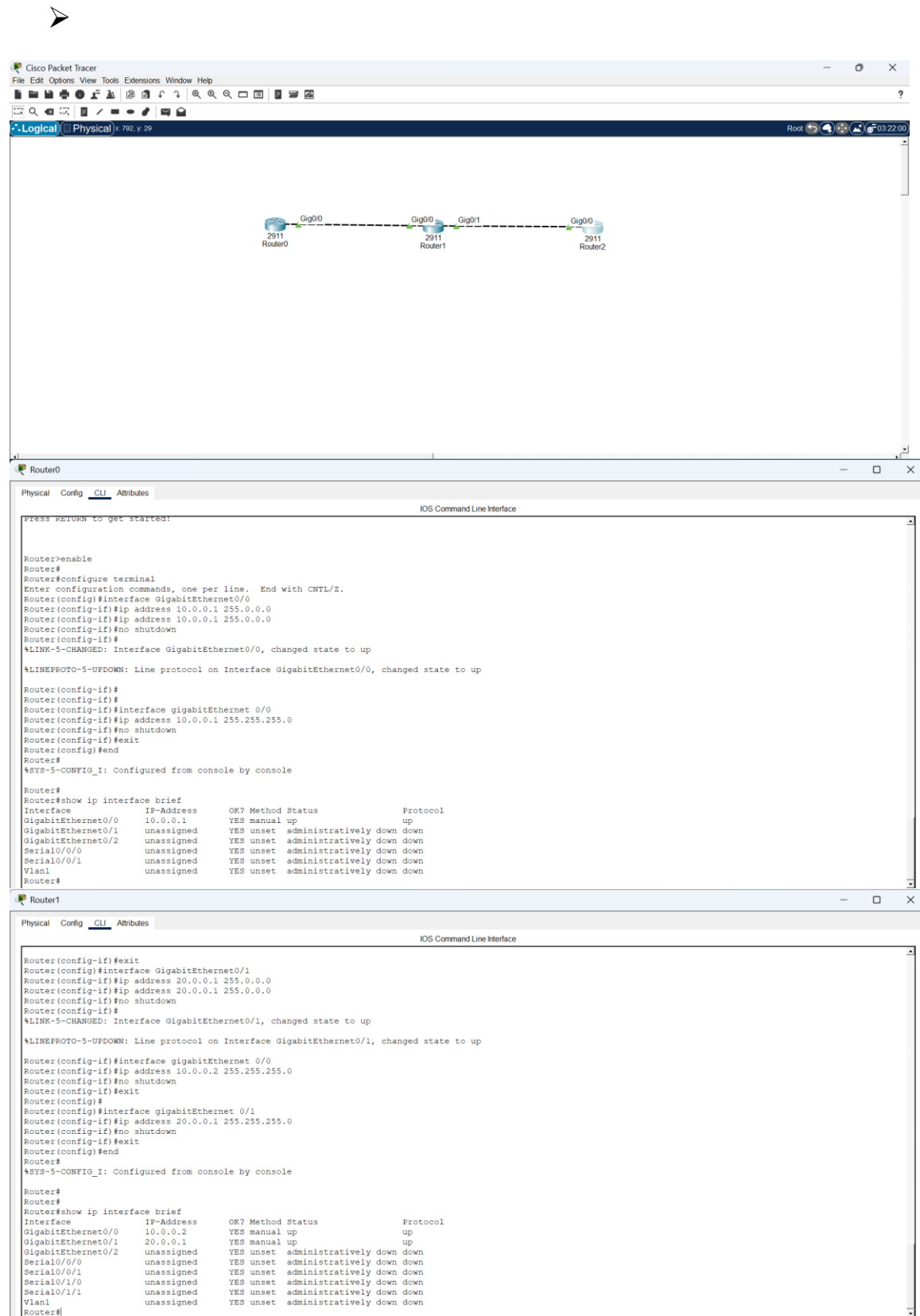


# 1. Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.



The screenshot displays the Cisco Packet Tracer interface. At the top, a network topology is shown with three routers (2911 Router0, 2911 Router1, and 2911 Router2) connected in a line. Router0 is connected to Router1 via Gig0/0 and Gig0/1, and Router1 is connected to Router2 via Gig0/0 and Gig0/1. Below the topology, the CLI windows for Router0 and Router1 are shown, displaying the configuration commands and the output of the 'show ip interface brief' command.

**Router0 CLI:**

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0  10.0.0.1        YES manual up            up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Serial0/0/0       unassigned      YES unset  administratively down down
Serial0/0/1       unassigned      YES unset  administratively down down
Vlan1            unassigned      YES unset  administratively down down
```

**Router1 CLI:**

```
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 20.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0  10.0.0.2        YES manual up            up
GigabitEthernet0/1  20.0.0.1        YES manual up            up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Serial0/0/0       unassigned      YES unset  administratively down down
Serial0/0/1       unassigned      YES unset  administratively down down
Serial0/1/0       unassigned      YES unset  administratively down down
Serial0/1/1       unassigned      YES unset  administratively down down
Vlan1            unassigned      YES unset  administratively down down
```

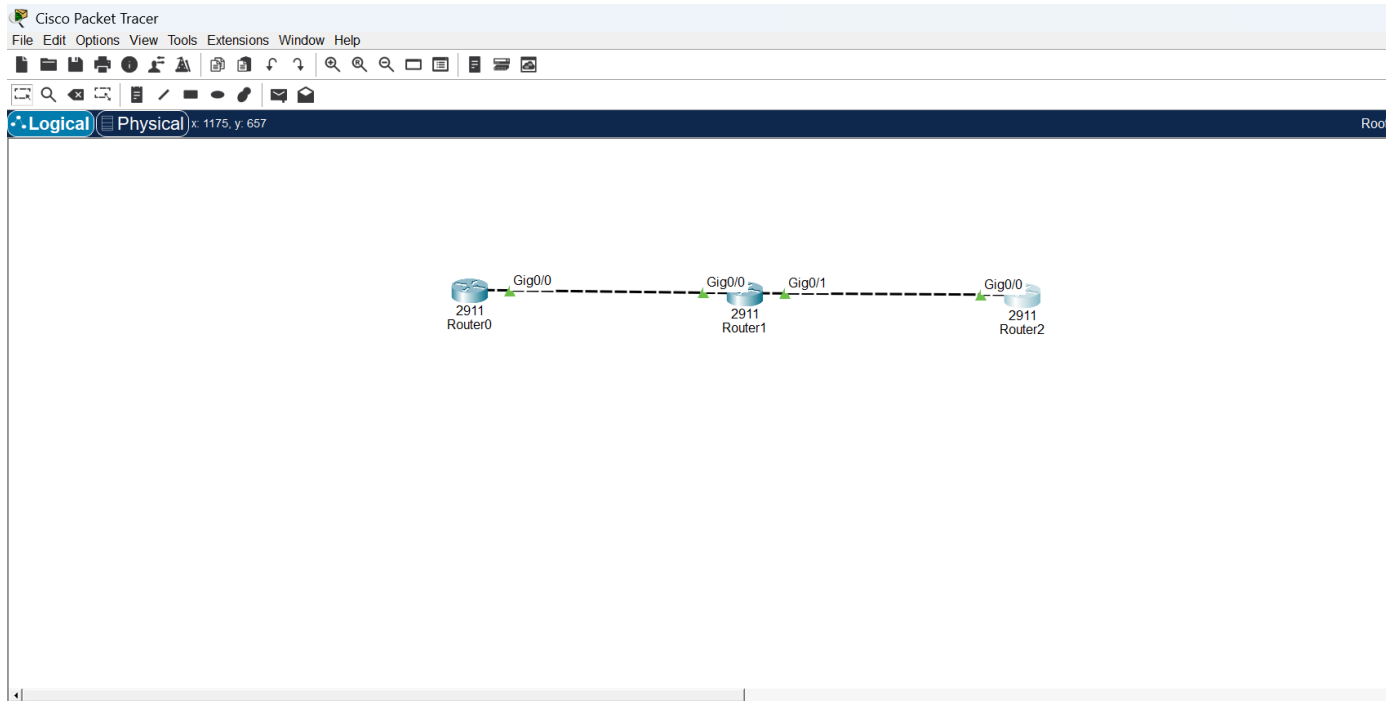
```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#interface gigabitEthernet 0/0
Router(config-if)#ip address 20.0.0.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  20.0.0.2        YES manual  up          up
GigabitEthernet0/1  unassigned      YES unset   administratively down down
GigabitEthernet0/2  unassigned      YES unset   administratively down down
Serial0/0/0        unassigned      YES unset   administratively down down
Serial0/0/1        unassigned      YES unset   administratively down down
Vlan1             unassigned      YES unset   administratively down down
Router#
```

**2.Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.**



```
Router0

Physical Config CLI Attributes

IOS Command Line Interface

GigabitEthernet0/1      unassigned      YES unset      administratively down down
GigabitEthernet0/2      unassigned      YES unset      administratively down down
Serial0/0/0             unassigned      YES unset      administratively down down
Serial0/0/1             unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       20.0.0.0/24 is directly connected, GigabitEthernet0/0
R       20.0.0.0/24 is subnetted, 1 subnets
        20.0.0.0/24 [120/1] via 10.0.0.2, 00:00:00, GigabitEthernet0/0
Router#
```

```
Router1

Physical Config CLI Attributes

IOS Command Line Interface

Serial0/1/0             unassigned      YES unset      administratively down down
Serial0/1/1             unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#network 20.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
Router#
```

```
Router2

Physical Config CLI Attributes

IOS Command Line Interface

Serial0/0/0             unassigned      YES unset      administratively down down
Serial0/0/1             unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 20.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 1 subnets
R       10.0.0.0/24 [120/1] via 20.0.0.1, 00:00:12, GigabitEthernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/0
L       20.0.0.2/32 is directly connected, GigabitEthernet0/0
Router#
```

### 3. Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

The image shows a Cisco Packet Tracer network diagram and the CLI configuration for Router3. The network diagram at the top shows four routers connected in a line: Router0 (2911) connected to Router1 (2911) via Gig0/0 and Gig0/1, Router1 connected to Router3 (2911) via Gig0/0 and Gig0/1, and Router3 connected to Router2 (2911) via Gig0/0 and Gig0/1. The CLI window for Router3 shows the configuration of interfaces g0/0 and g0/1 with IP addresses 20.0.0.2 and 30.0.0.1 respectively, and the output of the 'show ip interface brief' command.

```
Router3#  
Router(config-if)#ip address 30.0.0.1 255.0.0.0  
Router(config-if)#no shutdown  
Router(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up  
Router(config-if)#  
Router(config-if)#  
Router(config-if)#  
Router(config-if)#interface g0/0  
Router(config-if)#ip address 20.0.0.2 255.255.255.0  
Router(config-if)#no shutdown  
Router(config-if)#exit  
Router(config)#  
Router(config)#interface g0/1  
Router(config-if)#ip address 30.0.0.1 255.255.255.0  
Router(config-if)#no shutdown  
Router(config-if)#exit  
Router(config)#  
Router(config)#  
Router(config)#  
Router(config)#exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
Router#show ip interface brief  
Interface      IP-Address      OK? Method Status      Protocol  
GigabitEthernet0/0  20.0.0.2        YES manual up          up  
GigabitEthernet0/1  30.0.0.1        YES manual up          up  
GigabitEthernet0/2  unassigned      YES unset  administratively down down  
Serial0/0/0        unassigned      YES unset  administratively down down  
Serial0/0/1        unassigned      YES unset  administratively down down  
Serial0/1/0        unassigned      YES unset  administratively down down  
Serial0/1/1        unassigned      YES unset  administratively down down  
Vlan1            unassigned      YES unset  administratively down down  
Router#
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#
Router(config)#
Router(config)#show ip interface brief
% Invalid input detected at '^' marker.
Router(config)#
Router(config)#exit
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.0.0.1        YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Serial0/0/0        unassigned      YES unset  administratively down down
Serial0/0/1        unassigned      YES unset  administratively down down
Vlan1             unassigned      YES unset  administratively down down
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
Router#
```

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical 1175 y 657

Root 06:15:30

```
graph LR
    R0[2911 Router0] ---|Gig0/0| R1[2911 Router1]
    R1 ---|Gig0/1| R2[2911 Router2]
```

Time: 00:12:28 Realtime Simulation

**4. Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers.**  
**Hint:** Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.



The screenshot displays the Cisco Packet Tracer interface. The top section shows a network topology with four routers: Router0, Router1, Router3, and Router2. They are connected in a linear fashion: Router0 (Gig0/0) to Router1 (Gig0/1), Router1 (Gig0/1) to Router3 (Gig0/0), and Router3 (Gig0/0) to Router2 (Gig0/0). The bottom section shows the CLI of Router0, where the following commands have been entered:

```
Router>
Router>
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#end
Router#
Router#
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip route
```

